

Lab 1:

Objective: The goal of this lab is to reinforce the fundamental concepts of dynamic memory allocation, object-oriented programming, and deep copying in C++. Students will gain hands-on experience with constructors, destructors, copy constructors, and array-based dynamic memory management while developing a custom integer list class.

Problem Statement: In this lab, you are required to implement a dynamic integer list class in C++ that manages an array of integers.

Steps to follow:

1. Create a class for an integer list.
 - a. For example: [23,3,1,5,2,42,54,2,_,_] is a list with a capacity of 10 and it holds 8 values.
2. Data requirements
 - a. An integer array to hold the list
 - b. Its capacity
 - c. The current number of values it holds
3. Operational requirements
 - a. The basic constructor should dynamically allocate space to hold 10 numbers by default (unless specified otherwise) – capacity will be 10, size will be 0
 - b. A destructor to deallocate memory
 - c. Insert operation to add a value at the first free spot
 - d. If the vector is out of capacity it will not add a value and will indicate that it hasn't by returning a false (which means return a true when added, to indicate that we have added)
 - e. A copy constructor
 - f. A display function which prints an 8 element list with a capacity of 10 as [23,3,1,5,2,42,54,2,_,_]
 - g. Write a driver program to run tests to confirm that your copy constructor is working correctly